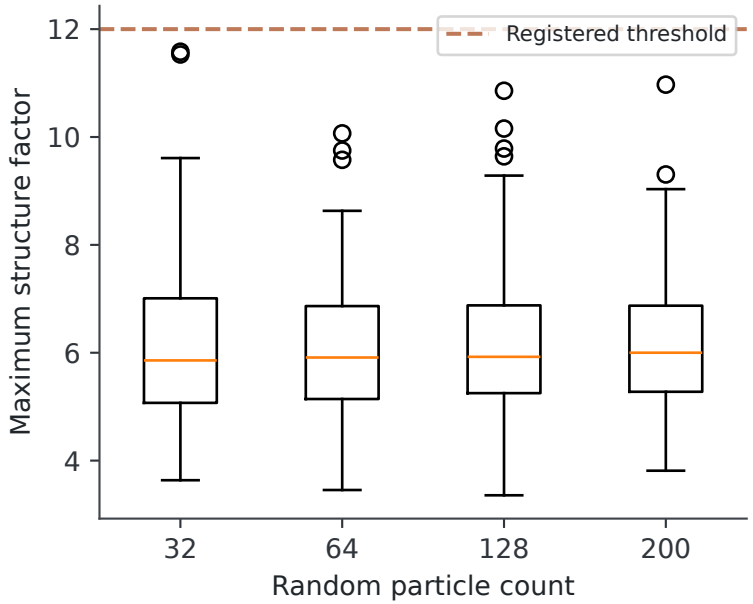
Spatial descriptors and false-positive controls



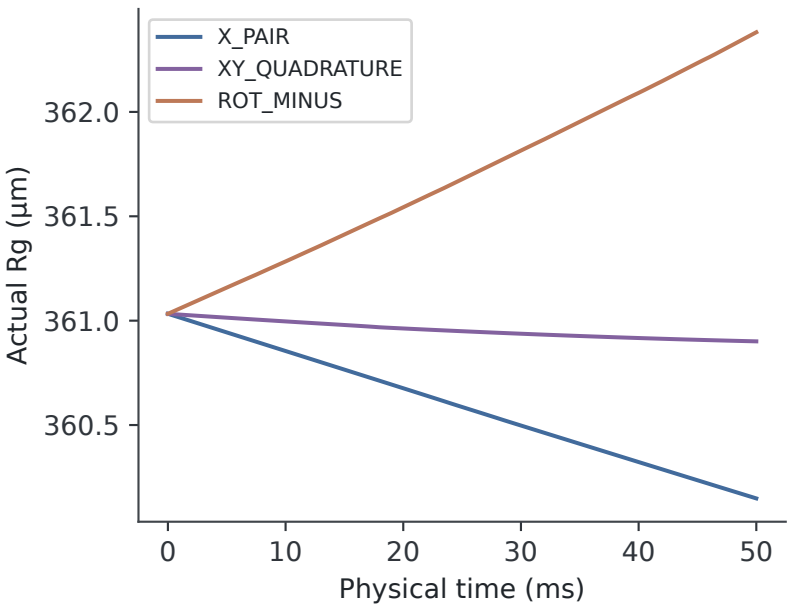
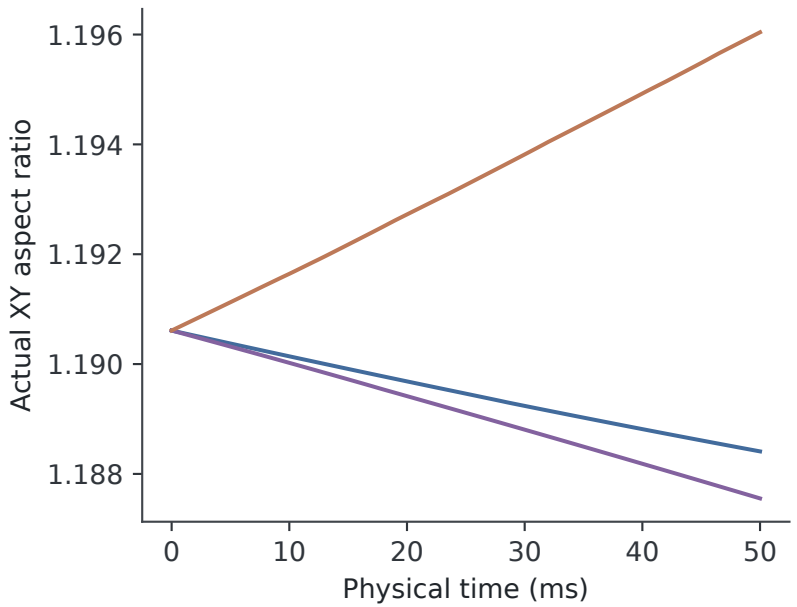
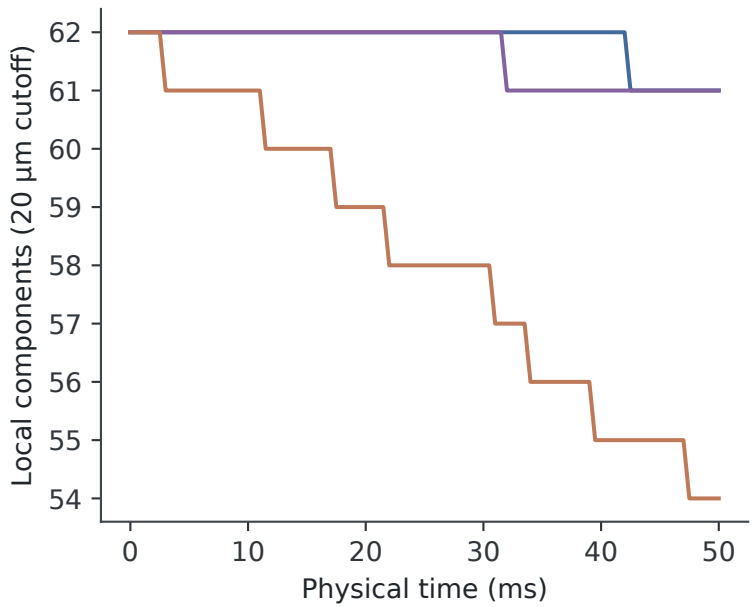
Metric unit tests: PASS

Uniform random clouds
Synthetic stripes and grid
Synthetic ring
Rigid rotation + translation
Pure shear negative control

Synthetic cases test analysis only.

F10Y distribution() reused unchanged

COM, Rg, covariance, aspect
Nearest neighbor and local density

Omega_z: center-motion rotation
omega_affine = A21 - A12
Planar rigid motion: omega = 2 Omega

No single-bubble spin state.